

# Guardians of the Forest

## Official Game Document

*Document Type: Post-Complete Game Design Overview (Version 1.0)*

*Project Status: Completed, June 2022*

### Elevator Pitch

'Guardians of the Forest' is a 2D fantasy puzzle-adventure game inspired by the core mechanics of the popular 'Angry Birds' franchise, where players fire magical projectiles from one of three unique wizards to destroy obstacles and defeat enemies, becoming a true Guardian of the Forest.

## Table of Contents

|                                |    |
|--------------------------------|----|
| Elevator Pitch                 | 1  |
| Table of Contents              | 2  |
| Game Concept                   | 3  |
| Game Summary                   | 3  |
| Genre & Target Audience        | 3  |
| Target Platform                | 3  |
| Unique Selling Points          | 4  |
| Story & Setting                | 4  |
| Premise                        | 4  |
| World & Environment            | 4  |
| Characters                     | 5  |
| Narrative Flow                 | 8  |
| Gameplay Overview              | 8  |
| Core Mechanics                 | 8  |
| Objectives                     | 8  |
| Rules & Systems                | 8  |
| Progression                    | 9  |
| Level Design                   | 10 |
| Level Layout                   | 10 |
| Challenges & Puzzles           | 10 |
| Rewards & Progression          | 10 |
| Art & Visuals                  | 11 |
| Art Direction                  | 11 |
| Characters & Environment       | 11 |
| HUD & UI Elements              | 13 |
| Animations                     | 16 |
| Audio                          | 18 |
| Music                          | 18 |
| Sound Effects                  | 18 |
| Controls & User Interface      | 18 |
| Controls                       | 18 |
| UI Flow                        | 19 |
| Technical Specifications       | 20 |
| Engine & Tools                 | 20 |
| Platform Requirements          | 20 |
| Performance Targets            | 20 |
| Development & Testing Hardware | 20 |
| Primary Development Machine    | 20 |
| Quality Assurance Testing      | 21 |
| Postmortem                     | 22 |
| Future Plans                   | 23 |
| References & Inspirations      | 24 |

## Game Concept

### *Game Summary*

*Guardians of the Forest* is a fantasy-themed, puzzle-adventure game where players select one of three wizards – each with their own unique projectile selection – to battle through eight enemy-filled levels to become a true Guardian of the Forest. Inspired by the “pull-and-launch” mechanics popularized by the *Angry Birds* franchise, the game reimagines this formula with magical spells, mystic interactive environments, and a fantasy family-friendly setting for all to enjoy.

As a first-time, solo-developed project, the primary goal was to gain hands-on experience with the full game development cycle while reinforcing game design knowledge and programming practices.

### *Genre & Target Audience*

*Guardians of the Forest* is a 2D single-player, puzzle-adventure game aimed at a wide and casual audience. Players who enjoy physics-based puzzle experiences, magical fantasy themes, or any title from the *Angry Birds* franchise will find the experience appealing.

Recommended age: **PEGI 3** – Mild violence.

### *Target Platform*

The game targets Windows PC exclusively. Limited experience with mobile and console development defined a scope centred on the Unity PC build pipeline.

### *Unique Selling Points*

- **Unique Take On A Classic Formula:** Mixes the classic pull-and-launch formula with a magical, fantasy twist.
- **Interesting Replayability:** With three unique wizards who each own two unique projectiles, there is plenty of replayability within the game.
- **Broad Target Audience:** The game is suitable for almost all ages with family-friendly visuals, gameplay, and themes.
- **Short But Sweet:** The small nature of the game ensures play sessions are satisfyingly short but sweet for players.

## Story & Setting

### *Premise*

The mystical forest of the Guardians has been seized by evil skeletons, goblins, and golems. As one of the three Guardian Wizards – each with their own unique magical projectiles to match their personality – you must push through enemy-occupied forest regions and reclaim the land, becoming a true Guardian of the Forest in the process. The game purposefully features a lightweight narrative to deliver an easy and enjoyable experience for the target audience without too much exposition.

### *World & Environment*

The game takes place across various mystical forest environments, each themed to match the encountered enemies. The tone is light, colourful, and accessible, blending whimsical fantasy with mild elements of danger. Every two levels represent a new section of the forest, showcasing the player's journey as they enter more dangerous and difficult areas.



Figure 1: The world environment of the first level of Guardians of the Forest.

## Characters

### Guardian Wizards (Protagonists)

#### Elemental Wizard

Bio: Focuses on bending Earthen elements to their will to destroy their enemies.

Projectiles: **Fireball**, **Electricball**

Playstyle: High-impact, destructive spell casting with visually explosive power.



Figure 2: The Elemental Wizard.

### *Healer Wizard*

Bio: Focuses on their medicinal skills to defeat enemies standing in their way.

Projectiles: **Bloodball, Poisonball**

Playstyle: Pre-planned, delayed interaction-based spell casting with devastating impact.



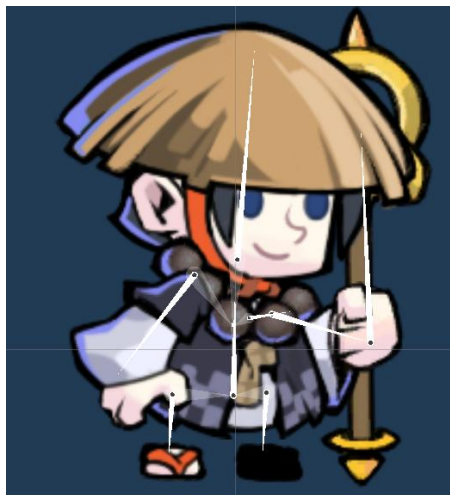
*Figure 3: The Healer Wizard.*

### *Monk Wizard*

Bio: Focuses on ancient, unhinged magic to perform heroic dark arts to overcome their enemies.

Projectiles: **Cosmicball, Voodoooball**

Playstyle: Strong, mystical spell casting with tide-turning capabilities.



*Figure 4: The Monk Wizard.*

## Enemies (Antagonists)

The forest is defended by three different enemies: **Skeletons**, **Goblins**, and **Golems**, each with three visual variants. The enemies function as key destructible targets rather than active threats, shifting gameplay focus towards environmental problem-solving and physics-based puzzle design.



Figure 5: The Wizard Skeleton, one of three Skeleton variants.

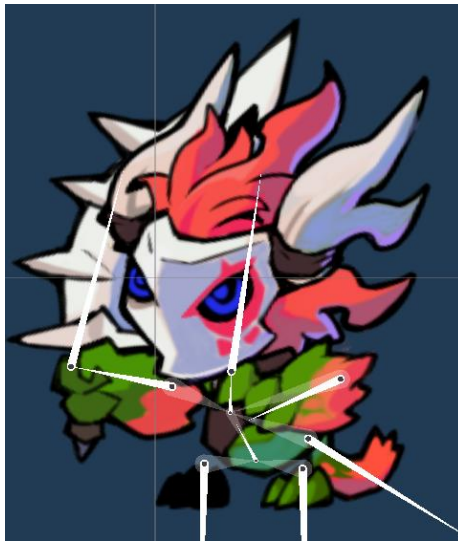


Figure 6: The Berserker Goblin, one of three Goblin variants.



Figure 7: The Ice Golem, one of three Golem variants.

### *Narrative Flow*

The story is narratively delivered subtly through level progression, with each advancing level representing a deeper section of the forest where enemy numbers gradually increase. Upon completion of the final eighth level, players are rewarded with an in-game certificate officially recognizing them as a Guardian of the Forest to end the story.

## Gameplay Overview

### *Core Mechanics*

Players click and drag their chosen wizard to aim their projectile before releasing the mouse button to fire away at the environment. Precision, timing, persistence, and projectile selection are key to success. Players utilize the following core actions:

- **Movement & Aiming:** To move the wizard and aim the projectile.
- **Firing:** To fire the projectile.
- **Activating Projectile Power-Up:** To activate the chosen projectile's Power-Up.

### *Objectives*

The one and only objective is to defeat all eight levels of the game. To do so, players must defeat all enemies within the level to advance. To defeat an enemy, players must either hit them directly with their projectile or knock an obstacle onto the enemy's head. The game gradually increases difficulty by introducing more enemies in the later levels with only a short increase in projectile count.

### *Rules & Systems*

Players are limited to three projectiles for the first six levels, and then five projectiles for the final two levels to combat the extra added enemies. Each projectile has its own unique Power-Up that can be activated once per level to aid players in their objective. If players do not defeat all enemies once all their projectiles have been used, they must restart the level. Below is a outlining each projectile in more detail.

| Projectile | Projectile Behaviour                                                                 | Power-Up Name | Power-Up Ability                                                              |
|------------|--------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------|
| Bloodball  | 'Bloodies' all objects it touches for its Power-Up combo with average size and speed | Bloodburst    | Bursts all bloodied objects at once, sending them flying in random directions |



|              |                                                                                     |                    |                                                                                                          |
|--------------|-------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------|
| Cosmicball   | Super-fast projectile with small size                                               | Cosmic Energy      | Enlarges the Cosmicball's size while retaining its super speed                                           |
| Electricball | Zaps all objects it touches with its big size making up for its slow speed          | Zap-Tastic!        | Triples the zapping power when electrocuting objects                                                     |
| Fireball     | Burns through all objects it touches with high speed                                | Laser-Beam of Fire | Fires a deadly laser-beam vertically, destroying everything in its path                                  |
| Poisonball   | 'Poisons' all objects it touches for its Power-Up combo with average size and speed | Poison Smoke       | Destroys all poisoned objects at once                                                                    |
| Voodoooball  | 'Curses' all objects it touches for its Power-Up combo with average size and speed  | Way of the Voodoo  | Destroys all cursed objects at once with a big explosion and randomly auto-kills a remaining alive enemy |

### *Progression*

Players enter deeper into the forest to continue their mission of clearing all enemies as they progress further. The final two levels of the game feature a higher enemy count, but reward players with an additional two projectiles to combat them. Early game builds also increased the projectile's Power-Up count to two for these final levels, but testing revealed that decision to cause the final levels to become too easy.

## Level Design

### Level Layout

Each level features a standing platform for the player wizard to stand and move on towards the left-hand side of the screen, while the right-hand side is filled with varied platform designs to hold enemies and obstacles. The environmental structures, enemies, and enemy count vary between levels, while every two levels feature its own unique background and style to represent a different region of the forest.



Figure 8: The level design of Level 4, featuring the player wizard movement platform to the left and the remainder of the level with obstacles and enemies to the right.

### Challenges & Puzzles

While the core challenge of the game is physics-based puzzles to destroy enemies and obstacles alike, the game's level design carefully positions these enemies and obstacles to encourage strategic projectile choice and aim, as well as smart Power-Up usage. Some projectiles excel in tight spaces, others in destruction, and others in open angles. This design encourages persistence, replayability, and smart thinking to discover the best solutions to defeat each level.

### Rewards & Progression

The further players advance, the more visual storytelling they uncover, as well as an additional two extra projectiles for the final two levels to combat the increased enemy count. Upon completion of the game, players are rewarded with a certificate congratulating them on becoming a true Guardian of the Forest.

## Art & Visuals

## Art Direction

The art direction is a stylized, colourful, cartoon aesthetic which complements the magical and family-friendly theme of the game. This decision allows for approachable visuals that bring the whimsical world to life. There were a handful of appropriate art directions for *Guardians of the Forest*, including retro pixel art, realism, and cel-shading, but the cartoonish, stylized look of the game's main inspiration, *Angry Birds*, led to the decision for this art style to be used, as it additionally offered great freedom over the creation of the magical world needed for the game's setting.



Figure 9: The world environment of Angry Birds, which inspired Guardians of the Forest.  
TheGamer (2024). Available at: <https://thegamer.com/angry-birds-friends-level-60-walkthrough-guide/>  
(Accessed 19<sup>th</sup> March 2026)

## Characters & Environment

Levels feature unique colours and atmospheres tailored to each specific forest region, showcasing various different aesthetics and environments within the world. All three wizards and enemies utilize the same cartoon approach to their aesthetic to easily merge into the world, with unique designs to showcase their personality traits for visual separation. The environmental background objects and obstacles further enhance the magical forest by seamlessly fitting into the created world with visuals to match, bringing the world to life with additional details.





Figure 10: Magical forest concept art inspiration.  
ArtStation (2016). Available at: <https://artstation.com/artwork/50qNz>  
(Accessed 19<sup>th</sup> March 2026)



Figure 11: More magical forest concept art inspiration.  
ArtStation (2021). Available at: <https://artstation.com/artwork/Oo4P5g>  
(Accessed 19<sup>th</sup> March 2026)

### HUD & UI Elements

The HUD features elements at the top and bottom of the screen to help players with their gameplay. The elements are appropriately sized to ensure gameplay is not affected negatively. These elements include:

- **Pause/Menu Button (Top-Left):** Pauses the game, displaying the Pause Menu.
- **Current Level Indicator (Top-Right):** Showcases the current level the player is on.
- **Selected Wizard (Bottom-Far Left):** Displays the player's selected wizard.
- **Selected Projectile (Bottom-Middle Left):** Displays the player's selected projectile.
- **Power-Up Activation Status (Bottom-Close Left):** Showcases the projectile's Power-Up activation status. The Power-Up is still available to use if this icon is not greyed out.
- **Remaining Enemies (Bottom-Right):** Shows the remaining enemies alive in the level.



Figure 12: The HUD.

The Main Menu is displayed on a hanging canvas in front of a region of the forest, displaying a variety of elements like:

- **Start Button:** For starting a new game.
- **Continue Button:** For continuing from a saved game.
- **Options Button:** For viewing and selecting the wizard and projectile options.
- **Credits Button:** For viewing the credits of the game.
- **Quit Button:** For quitting the game.



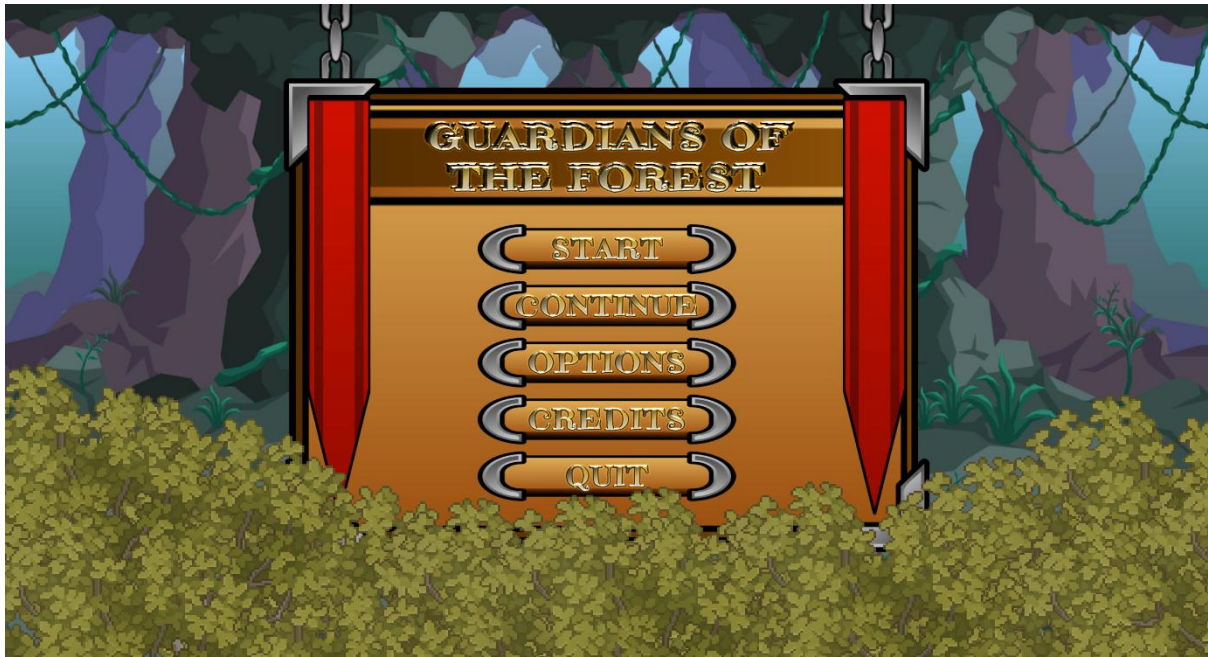


Figure 13: The Main Menu.

The Pause Menu utilizes the same hanging canvas design as the Main Menu, just smaller to partially take up space on the gameplay screen. Elements of this menu include:

- **Resume Button:** Resumes the game.
- **Options Button:** Same function as the Main Menu, minus the ability to select a new wizard and/or projectile.
- **Quit Button:** Quits to the Main Menu.



Figure 14: The Pause Menu.

The Level Failed Menu is similar in design to the Pause Menu, but features just the one element:

- **Continue Button:** Restarts the failed level.

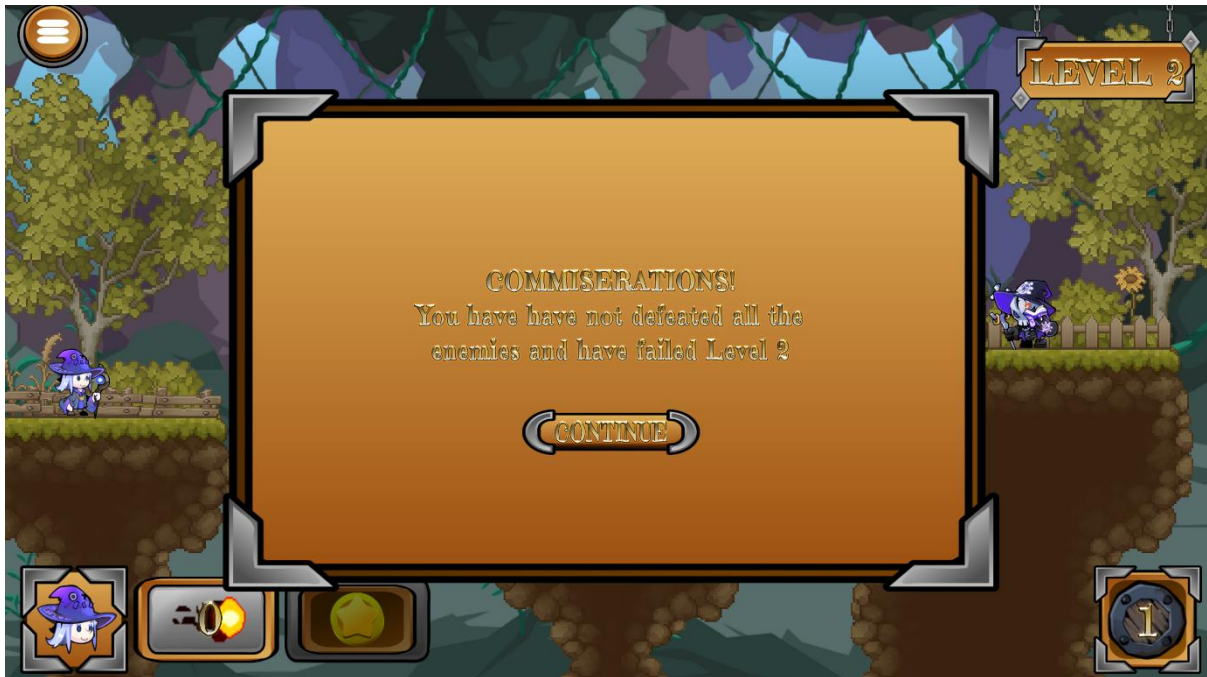


Figure 15: The Level Failed Menu.

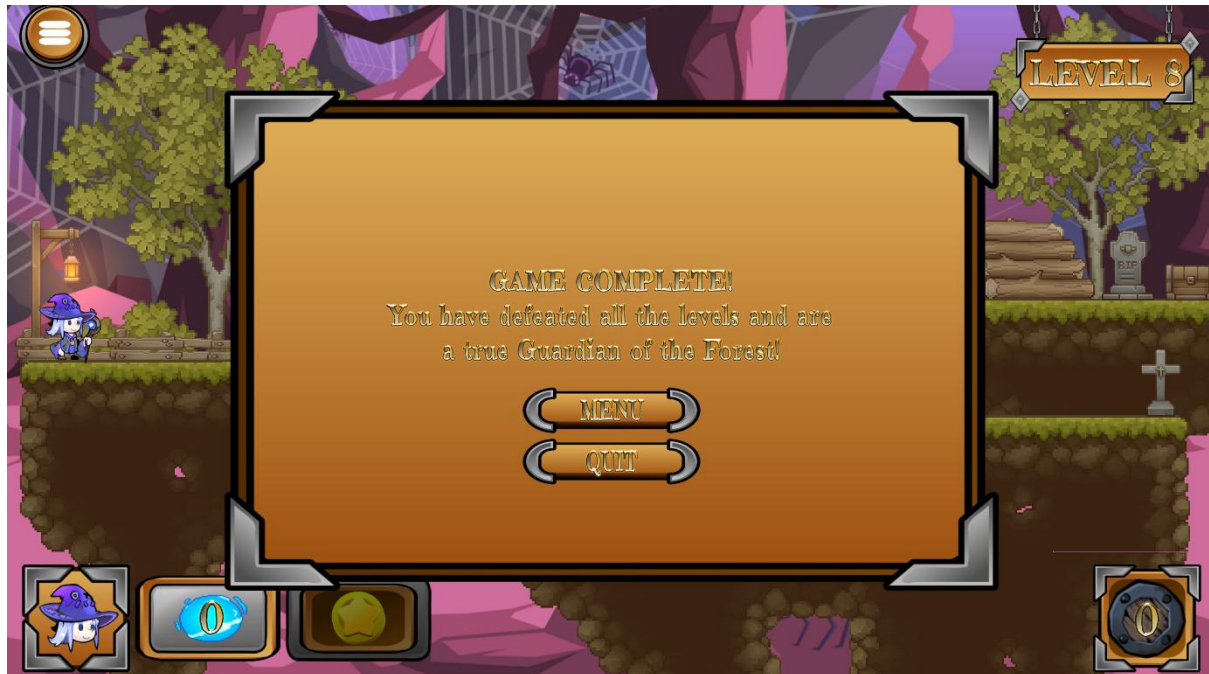
The Level Completed Menu is the same as the Level Failed Menu but features different informational text and its Continue Button starts the next level instead of the current one.



Figure 16: The Level Completed Menu.

The Game Completed Menu also uses the same design as both the Level Failed Menu and Level Completed Menu, containing its own elements like:

- **Menu Button:** Returns to the Main Menu.
- **Quit Button:** Quits the game.



*Figure 17: The Game Completed Menu.*

The HUD and menus utilize the same one font which was carefully selected to match the theme of the game while also being accessible for all to read with ease.

## *Animations*

### **Player**

Each Guardian Wizard has animations for idling and firing projectiles, individually suited to their own personalities to ensure each not only has its own look and gameplay designs, but also its own identity through lifelike movements. Additionally, upon completion of a level, each wizard has its own victory celebration animation to celebrate.

### **Projectiles**

All six projectiles have their own animations for movement, collision, and Power-Up activation. Every one of these animations was carefully selected to bring each projectile to life with their own unique identities, acting as main characters themselves. For example, the Fireball uses a red laser-beam animation for its Power-Up, while the Poisonball uses a big green smoke poof for its Power-Up, distinctively separating each projectile while retaining its personality in its animation.



Some projectile Power-Up animations, such as the Fireball and Voodoooball, feature multiple visual effects merged into one for a grand design. Although the visuals were sourced from a designer, these specific multi-merge animations were created by myself by playing them over one another to create the design.

### **Enemies**

All three enemies have their own idling animations shared across their variants. All three enemies share the same death animations due to the functionality being shared and not requiring specific unique animations.

### **Environment**

The environment is full of animated background objects and obstacles, such as fragile wooden planks, boxes, chests, campfires, and more. These help bring the world and the game to life with a sense of realism, further helping to enhance the magical effect of the game.

### **UI**

All menus use a handcrafted animation which sees the hanging canvas drop down from the top of the screen on chains to add more magic and personality to the overall feel of the game.

## Audio

### Music

*Guardians of the Forest* is an upbeat, casual, fantasy themed game, and therefore uses appropriate, family-friendly, yet engaging soundtracks for each individual level and menu. Each level progressively gets harder as the player nears the more dangerous regions of the forest, which is not only shown by the scenery and enemy count, but also through the soundtracks which progressively get more intense.

The menus ensure players feel comfortable and relaxed with charming fantasy-themed music appropriate for menu navigation.

### Sound Effects

There are a handful of sound effects featured in *Guardians of the Forest* to further help bring the magical world to life. The wizards all share the same magical firing sound when a projectile is fired, and each enemy type uses its own idling sound to audibly show they are alive – until defeated, where a universally shared, distinct death and poof sound are played one after another for audible death confirmation.

All six projectiles have their own unique sounds for movement, collision, and Power-Up activation to provide important audio cues for players, while all UI elements utilize appropriate sound effects, such as button clicks, to create a polished design to ensure players feel as immersed as possible within the experience.

## Controls & User Interface

### Controls

- **Movement & Aiming (Left-Mouse Press):** Allows the player to move their character and aim the projectile.
- **Fire (Left-Mouse Release):** Allows the player to fire their projectile.
- **Activate Projectile Power-Up (Spacebar):** Allows the player to activate their projectile's Power-Up.

A key rebinding system was not implemented due to the casual, easy-to-play nature of the game and controls, which are based on the iconic pull-and-launch gameplay mechanic.

### *UI Flow*

The HUD can feature a handful of elements at any given time, but they are all strategically sized and placed to avoid interruption with gameplay. To ensure consistency across all HUD, menus, and other UI elements, the same 'hanging canvas' design is used, with matching colour schemes and the one same font. All menus support full mouse navigation for ease-of-use.

## Technical Specifications

### Engine & Tools

*Guardians of the Forest* was developed with Unity v2022, with C# scripting. Visual Studio 2022 (coding software) and Audacity (audio manipulation software) was also used in the creation of the game.

### Platform Requirements

|           | Minimum                                        | Recommended                                    |
|-----------|------------------------------------------------|------------------------------------------------|
| OS        | Windows 10 64-bit                              | Windows 10/11 64-bit                           |
| Processor | Intel Core i5-4 <sup>th</sup> Gen / equivalent | Intel Core i7-9 <sup>th</sup> Gen / equivalent |
| Memory    | 4 GB RAM                                       | 8 GB RAM                                       |
| Graphics  | NVIDIA GeForce GTX 750 Ti / AMD Radeon R7 260X | NVIDIA GeForce GTX 1060 / AMD Radeon RX 580    |
| VRAM      | 1 GB                                           | 3 GB                                           |
| Storage   | 200 MB available space                         | 500 MB available space (SSD)                   |
| DirectX   | Version 11                                     | Version 11                                     |

### Performance Targets

- **Frame Rate:** 60 FPS at 1080p on Minimum Specs, 120 FPS at 1080p on Recommended Specs
- **Resolution:** 1920x1080
- **Loading Times:** ~1-2 seconds on SSD; ~3-6 seconds on HDD

## Development & Testing Hardware

### Primary Development Machine

|           |                                     |
|-----------|-------------------------------------|
| OS        | Windows 10 64-bit                   |
| Processor | Intel Core i7-9750H                 |
| Memory    | 16 GB RAM                           |
| Graphics  | NVIDIA GeForce RTX 2080 Super Max-Q |
| VRAM      | 8 GB                                |
| DirectX   | Version 12                          |



### *Quality Assurance Testing*

The game underwent multiple rounds of playtesting during the middle and late production phases. These tests were conducted by myself and a few close others on the primary development machine. Testing focused on specific gameplay mechanics, overall fun and ease-of-use of the game, bugs and issues, and performance. These tests allowed for data to be gathered over the targeted testing areas, allowing changes to be made on any low-scoring sections before the game's full release.

## Postmortem

### What Went Well

- Many skills from learning the Unity game engine to programming practices not only with the C# language but also with general practices that can be used across various other languages were successfully learned throughout the project's creation – which was one of the main goals of the project.
- The game works as intended with smooth, enjoyable gameplay and a unique, fantasy theme bringing the entire project together.
- If the project was to be revisited for additional work, a solid foundation has been created to allow this through adaptable and scalable code.

### What Could Have Been Better

- The project began with minimal pre-production planning, with only the idea of the game being established without any proper plan to implement the project. Much of the project was made reactively rather than intentionally, which didn't affect the production much due to the game's small scope, but could have affected it more if the game was larger.
- The game was tested only on the primary development machine and by a small group of people, meaning the breadth of user testing was limited which therefore limited the amount of feedback received.
- The use of version control with Git was at times minimal and inconsistent due to the project being a first solo development and knowledge of version control was limited. Taking the time to learn more about this pre-production would have resulted in better version control management – which is crucial to a game's development cycle.
- Some features took a while to implement due to lack of experience. Better scheduling would have helped keep development momentum while managing expectations.

### Key Lessons Learned

- The importance of the pre-production phase of development is not to be underestimated. Creating documentation, prototypes, technical plans, and learning necessary systems beforehand dramatically improves efficiency and clarity – and reduces redesign work later.
- Building small and functional systems then iterating on them proved far more effective than building large interconnected systems all at once.
- Tracking workload and setting realistic goals keeps the project's scope small and achievable.

## Future Plans

*Guardians of the Forest* is a standalone experience with no plans for any downloadable content (DLC) packages or future sequels. However, the game could easily be expanded upon due to the solid framework created, where a variety of new levels, wizards, projectiles, and enemies could be implemented if any thoughts of a return were to occur.

## References & Inspirations

### Historical

The game's main mechanic, the pull-and-launch system, is a well-known, classic gameplay feature that has existed in many games over the years. *Angry Birds* is arguably the most popular video game choice for examples that utilize the mechanic, while a variety of golf themed games and mobile top-down games also employ the system.

### Games

The main inspiration for *Guardians of the Forest* is, of course, the *Angry Birds* franchise. The game uses the same core mechanics and gameplay themes, just with its own twist on setting, story, characters, and some minor mechanical changes. As mentioned earlier in the document, the decision to heavily base the foundation of the game on *Angry Birds* came from the fact that this is a first full solo project, so utilizing a well-known mechanic (such as pull-and-launch) seemed a smart idea to incorporate into the game.

### Other

Fantasy and magic themes are heavily present throughout the game. These ideas were chosen as they are widely regarded as family-friendly and help bring imagination to life in fun and creative ways with minimal restrictions. Due to the game being largely based off *Angry Birds*, the setting and theme of the game needed a change to create its unique identity, separating itself from its inspiration.